V-pairs Have the Almost Weak Maximizing Property

Candidate partial result packet for arXiv:2301.05003

Run `fa_banach_001`; prepared 2026-07-01

Source Question

The source paper is Mingu Jung, Gonzalo Martínez-Cervantes, and Abraham Rueda Zoca, *Rank-one perturbations and norm-attaining operators*, arXiv:2301.05003. In Question 3.7 the authors ask:

> Does every V-pair have the WMP?

The same page explains why an operator-level approach is insufficient: there are V-operators with non-weakly-null maximizing sequences that do not attain their norms, although the pair in that example is not a V-pair.

> Note that the pair (ℓ_p, ℓ_q) with $1 < p < q < \infty$ has the WMP (and therefore the CPP) while it is not a V-pair. We do not know whether the reverse implication holds for every pair of Banach spaces:
>
> **Question 3.7.** *Does every V-pair have the WMP?*
>
> A natural idea to prove that every V-pair has the WMP would be to try to show that every V-operator with a non-weakly null maximizing sequence is norm-attaining. However, the following remark shows that there are V-operators with non-weakly null maximizing sequences which do not attain their norm.

Definitions

A sequence $(x_n) \subset B_X$ is a maximizing sequence for $T \in \mathcal{L}(X, Y)$ if $\|Tx_n\| \to \|T\|$. A pair (X, Y) has the weak maximizing property, or WMP, if every operator $T : X \to Y$ with a non-weakly-null maximizing sequence attains its norm.

For this packet, say that (X, Y) has the almost weak maximizing property, or aWMP, if every operator $T : X \to Y$ for which every maximizing sequence is non-weakly-null attains its norm. Thus WMP implies aWMP, but aWMP is weaker: it only detects operators whose maximizing sequences are all non-weakly-null.

We use the standard approximate fixed-point formulation of a V-operator used in Khatskevich–Ostrovskii–Shulman and in Remark 3.8 of the source paper: if $\|T\| = 1$ and T is a V-operator, then there are $S \in \mathcal{L}(Y, X)$ with $\|S\| = 1$ and $y_n \in S_Y$ such that

$$\|TSy_n - y_n\| \longrightarrow 0.$$

Also, the domain of a V-pair is reflexive; equivalently, every bounded sequence in X has a weakly convergent subsequence.

Claimed Partial Result

Theorem 1. *Every V-pair has the almost weak maximizing property.*

Proof Intuition

Let $T : X \to Y$ be a norm-one operator on a V-pair. The V-pair assumption supplies an almost fixed-point sequence y_n for TS. Then Sy_n is a maximizing sequence for T. If every maximizing sequence for T is non-weakly-null, reflexivity lets us pass to a subsequence $Sy_n \rightharpoonup x_0 \neq 0$. The almost fixed-point relation forces $y_n \rightharpoonup Tx_0$ and hence $STx_0 = x_0$. Since $\|S\| = \|T\| = 1$, this identity can only happen if $\|Tx_0\| = \|x_0\|$, so T attains its norm.

Proof

Let (X, Y) be a V-pair and let $T \in \mathcal{L}(X, Y)$ be nonzero. Normalize so that $\|T\| = 1$, and assume that every maximizing sequence for T is non-weakly-null.

By the V-pair property and the approximate fixed-point formulation above, there are $S \in \mathcal{L}(Y, X)$, $\|S\| = 1$, and $y_n \in S_Y$ such that

$$\|TSy_n - y_n\| \to 0.$$

It follows that

$$\|TSy_n\| \to 1.$$

Since $\|Sy_n\| \leq 1$, the sequence $(Sy_n) \subset B_X$ is a maximizing sequence for T. By hypothesis, (Sy_n) is not weakly null.

The domain X is reflexive. Hence, after passing to a subsequence, we may assume that $Sy_n \rightharpoonup x_0$ for some $x_0 \in B_X$. Because the sequence is not weakly null, choose the subsequence so that $x_0 \neq 0$. Then

$$TSy_n \rightharpoonup Tx_0.$$

The norm convergence $TSy_n - y_n \to 0$ gives $y_n \rightharpoonup Tx_0$. Applying S, we also have $Sy_n \rightharpoonup STx_0$. Weak limits are unique, so

$$STx_0 = x_0.$$

Therefore

$$\|x_0\| = \|STx_0\| \leq \|Tx_0\| \leq \|x_0\|.$$

Thus $\|Tx_0\| = \|x_0\|$, and $x_0 \neq 0$. Consequently

$$\left\| T \frac{x_0}{\|x_0\|} \right\| = 1 = \|T\|,$$

so T attains its norm.

Consequence for Question 3.7

Corollary 1. *If a V-pair (X, Y) fails the weak maximizing property, then there is a non-norm-attaining operator $T : X \to Y$ with both of the following features:*

(i) *T has a non-weakly-null maximizing sequence;*

(ii) *T also has a weakly-null maximizing sequence.*

Proof. Failure of WMP gives a non-norm-attaining T with a non-weakly-null maximizing sequence. The theorem says that T cannot have all maximizing sequences non-weakly-null. Hence at least one maximizing sequence for T is weakly null. □

Limitations and Novelty Check

This is only a reduction, not a solution of Question 3.7. The remaining problem is exactly the mixed-sequence case in the corollary.

Cheap duplicate checks were run against the run registry, solution, attempt, and proof-gap indexes for arXiv:2301.05003 and for the core phrases aWMP, "almost weak maximizing", "V-pair", and "weak maximizing property". No existing packet for arXiv:2301.05003 or this reduction was found. A local later-source search found newer WMP sufficient conditions, in particular via strict property (M), but did not find an answer to the exact V-pair question. Web searches for the exact phrase "Does every V-pair have the WMP" and nearby variants returned no exact later answer.

Important provenance caution: the arXiv source TeX for arXiv:2301.05003 contains a commented-out draft of essentially the aWMP observation. The published PDF does not state it. This packet is therefore best viewed as a precise promoted cleanup and queue-memory record with modest novelty.

References

1. S. Dantas, M. Jung, and G. Martínez-Cervantes, *Some remarks on the weak maximizing property*, J. Math. Anal. Appl. 504 (2021), Article 125433.

2. M. Jung, G. Martínez-Cervantes, and A. Rueda Zoca, *Rank-one perturbations and norm-attaining operators*, arXiv:2301.05003.

3. V. A. Khatskevich, M. I. Ostrovskii, and V. S. Shulman, *Extremal problems for operators in Banach spaces arising in the study of linear operator pencils*, Integral Equations Operator Theory 51 (2005), 109–119.

4. M. I. Ostrovskii, *Extremal problems for operators in Banach spaces arising in the study of linear operator pencils, II*, Integral Equations Operator Theory 51 (2005), 553–564.

Human Review Recommendation

Review as a likely valid partial reduction. The main convention to verify is the approximate fixed-point form of V-operators for the real-space definition used in the source paper. Even if accepted, the packet should not be counted as a full solution to Question 3.7.